

A Novel Sign Language Translator : HandSpeak

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Abstract— Speech-impaired people's communication barriers with the rest of the public, which usually result in social and logistical issues. As we know, clear communication plays a vital role in saving lives in case of emergencies. The only way a Speech challenged person can communicate is through sign language. Sign language, however efficient, is not everyone's language. Therefore the proposed 'HandSpeak' system would fill that gap by creating a sign language translator glove with an Arduino Mega 2560, five flex sensors, and an I2C module. The system records finger movements equivalent to sign language movements and converts them into readable text on an LCD or via serial communication. A flex sensor is fixed to each of the fingers to sense the bending level, which is processed by the Arduino. The I2C module enables effective communication with peripherals such as displays. The device is low cost and easy to use, and has possible applications for assistive technology, education, and communication access.

Keywords—Sign Language, Arduino, Smart Glove and Flex Sensor.

I.INTRODUCTION

Communication is an essential part of human interaction and plays a vital role in society. While verbal communication is the most prevalent, many of the population practice non-verbal communication, specifically sign language, as a result of hearing or speech impairment. Sign language is not one that is used worldwide, however, resulting in gaps in communication between the hearing-impaired and non-practitioners of this type of language [1]. The main objective of 'HandSpeak' is to assist effective communication between people who use sign language and the people who doesn't. It also aim fill this gap through an embedded system that converts sign language gestures into readable text in real-time.

Sign language employs hand movements, body position, and facial expression to communicate meaning. For most deaf and mute people, it is the primary means of communication. Sadly, a lack of general awareness about sign language

among the public generally leads to problems for the speech- and hearing-impaired population [2]. This limitation encourages the creation of a system that is capable of interpreting sign language gestures and translating them into a more widely understood form—text.

The system developed in this paper employs an Arduino Mega 2560, a strong and highly versatile microcontroller platform that comes with a high number of input/output pins, which is perfect for managing several sensors at once. Finger movement is sensed using flex sensors placed on a glove, one sensor per finger. The sensors vary in resistance depending on how bent they are. The analog signals of the flex sensors are inputted into the Arduino Mega, where the signals are processed and the gesture made is ascertained [3].

In order to make communication between the Arduino and display elements (like an LCD) simpler, an I2C (Inter-Integrated Circuit) module is employed. I2C is a serial communication protocol in which several slave devices can be connected to a single master through just two wires: SDA (data line) and SCL (clock line). This simplifies wiring and makes it possible to have a neater and more organized design [4].

The system as proposed is wearable in the form of a glove, hence portable and convenient. It identifies pre-programmed gestures for letters or words and translates the equivalent text on an LCD display. Not only does it enable real-time communication, but it also supports learning and accessibility [5].

II.RELATED WORK

Various research studies and projects have attempted to translate sign language to text or speech via sensor-based systems. One such popular method utilizes gloves integrated with multiple flex sensors, one sensor attached to a finger to measure bending movement. Each movement translates into a corresponding letter or digit, and sensor information is interpreted to identify static hand signs. Most initial applications employed flex sensor gloves attached to

microcontrollers to identify simple American Sign Language (ASL) hand movements and display the findings on screens.

Arduino-based systems have become increasingly popular in gesture recognition applications because of their low cost, simplicity of programming, and extensive community base. For example, applications such as "Smart Glove for Sign Language Recognition" used an Arduino Mega together with flex sensors and accelerometers to identify finger positions and orientation of the hand. The sensor readings were processed to recognize gestures, and the results were displayed on an LCD screen or wirelessly transmitted.

Flex sensors are a crucial part of these systems due to their ability to alter resistance with bending, offering a simple yet useful means of sensing finger movement. Calibrating flex sensors to work for multiple angles, prior research ensured reliable gesture recognition using very little hardware complexity. Flex sensors are particularly ideal for static gesture translation where each hand pose is mapped to one alphabet or digit.

Also, I2C communication for LCDs is now a staple for such projects. I2C LCDs simplify wiring and save microcontroller pins significantly, leading to the overall system design being more efficient and streamlined. Such projects as the "Wireless Sign Language Translator" employed I2C LCDs in order to visually show translated words and letters in real-time, enhancing user interaction.

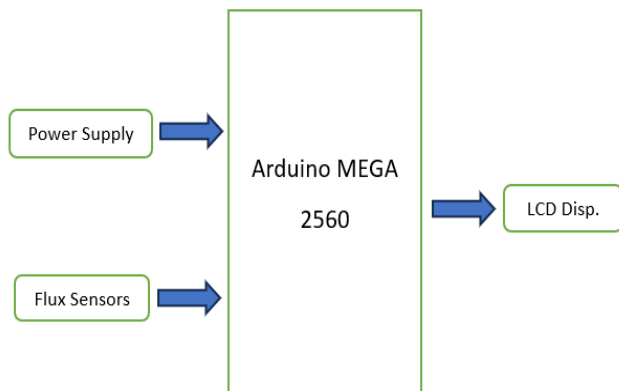


Fig 1: Block diagram

Even with these developments, most current systems are limited. They tend to deal with static hand signs and not dynamic gestures or complete word translations. Most systems are restricted to recognizing single letters, and hence full sentence construction is difficult. Nevertheless, these previous works give a solid foundation for creating better, more user-friendly sign language translators with current microcontrollers and sensor technologies.

III .PROPOSED SYSTEM

The HandSpeak suggested here is a wearable device that interprets sign language gestures into readable text through an Arduino Mega 2560, five flex sensors, and an I2C LCD display. Flex sensors are placed on every finger of a glove to sense bending [6]. When the user performs a hand gesture, the flex sensors alter resistance, producing analog signals that are captured by the Arduino. Dependent on the flex pattern of

the fingers, Arduino recognizes the resultant alphabet or letter [7].

Arduino Mega 2560 is chosen due to its large number of input pins and robust processing power, which can process multiple sensor inputs effectively [7]. The predefined threshold values are utilized to translate sensor values into particular letters [8]. The output character is then shown on an I2C LCD, which uses only two lines of communication and makes wiring easy [9].

This system is primarily intended to identify static gestures that symbolize single letters of the alphabet [10]. It provides an inexpensive, user-friendly solution that ensures improved communication for people who depend on sign language [11]. In the future, the system can be further developed to identify full words or dynamic gestures by incorporating additional sensors or machine learning models [13]. Generally, the device is a practical and mobile solution for bridging the hearing-impaired communication gap in the hearing- and speech-impaired community .

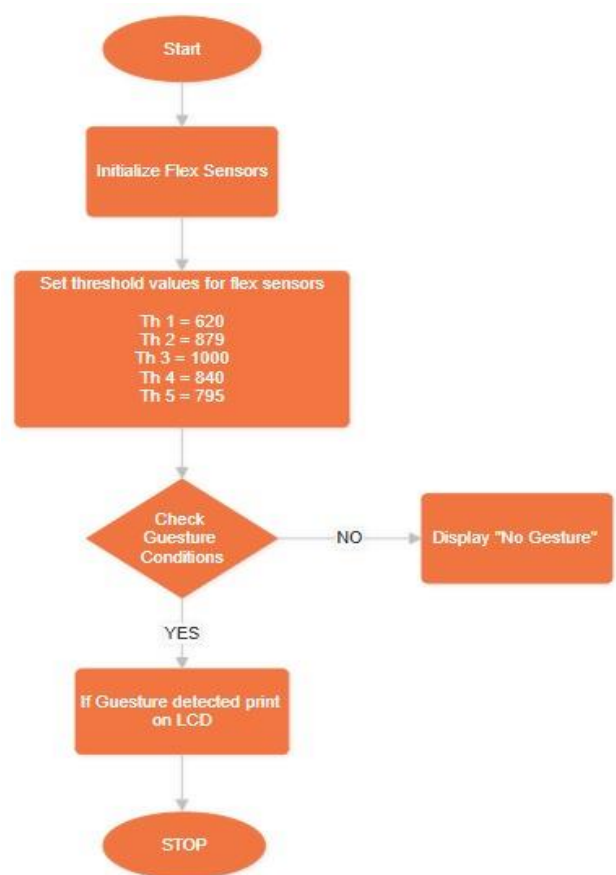


Fig 2: Flow chart

IV. IMPLEMENTATION SETUP

A. COMPONENTS REQUIRED

Hard Ware requirements:

1. Arduino Mega 2560
2. Flex Sensors (5 pieces)
- 3.I2C LCD Display (16x2)
- 4.Resistors-5(10K Ω)
- 5.Wires and Breadboard
- 6.Power Supply (USB or External)
- 7.Glove (or wearable material)
- 8.Jumper Wires (Male to Female)
- 9.Connectors

Software requirements:

- 1.Arduino IDE

1. Arduino Mega(2560) :

The Arduino Mega 2560 is a high-performance microcontroller board powered by the ATmega2560 chip, used for advanced embedded projects. It has 54 digital I/O pins, 16 analog inputs, and several communication interfaces such as SPI, I2C, and UART, suitable for projects that need lots of sensors and actuators [12]. The board contains 256 KB of flash memory, 8 KB of SRAM, and runs at a clock speed of 16 MHz [13]. Due to its ample memory, multiple input/output pins, and sophisticated communication features, the Arduino Mega 2560 is ideal for use in robotics, IoT projects, and wearable technology [14]. It is easily programmable with the Arduino IDE and is commonly used for projects involving extensive data processing and several devices [15].

2.Flex sensor:

A flex sensor is an electronic sensor that measures the degree of bending or flexing of an object by varying its resistance. The resistance of the sensor increases as it is bent, and this variation is used to measure the degree of bending. Flex sensors are usually fabricated from a thin, flexible material. Flex sensors are commonly implemented in voltage divider circuits, where the variation in resistance is measured by a microcontroller such as Arduino. These sensors find extensive use in gesture recognition, robotics, and wearable technology applications, where minute movement detection is crucial, like monitoring finger movement in sign language interpretation systems. They are best suited for interactive and motion-tracking technologies due to their simplicity, bending sensitivity, and flexibility [15].



Fig 3: Flex Sensor

3.I2C LCD Display(16x2):

The I2C LCD display is a small and effective display module that employs the use of the I2C (Inter-Integrated Circuit) protocol to exchange information with microcontrollers such as Arduino. It has four pins in general: VCC for supply power, GND for ground, SDA (Serial Data) for data transfer purposes, and SCL (Serial Clock) for clock synchronization. These two communication lines, SDA and SCL, minimize the number of needed pins over standard LCD modules, which makes it perfect for projects where there are limited I/O resources. The module generally has a 16x2 or 20x4 character display, permitting several lines of text to be displayed. The I2C LCD module makes programming and wiring easy because data and control commands can be transmitted using the I2C protocol, and it is ideal for embedded systems and uses such as sensor readings, user input, and monitoring systems. Its simplicity of integration, lower wiring complexity, and compact nature make it widely used in most electronics projects [16].



Fig 4: I2C LCD(16x2) display

4. Glove:

A glove is a wearable component that covers the hand and usually has distinct sections for every finger. For projects involving detection of hand movement or interaction, gloves are used frequently in the context of electronics and wearable technology. For instance, in sign language translation devices, gloves are provided with sensors (such as flex sensors) to sense finger bending and hand movement. Such gloves can be engineered to feed actual-time data into a microcontroller, which reads the data and translates it into output, for example, text or voice. Gloves are applied extensively in virtual reality (VR), robotics, gesture recognition, and other areas, allowing natural and hand-free control by sensing finger movement and other gestures. They are flexible and versatile, and hence, useful tools in interactive systems .



Fig 5: Glove

V. WORKING PRINCIPLE

The sign language translation glove works by sensing hand gestures through the use of flex sensors mounted on a wearable glove. The sensors are resistive bend sensors whose resistance varies with bending. There is one sensor per finger mounted, and it gives analog voltage outputs corresponding to the amount of bending of a finger. While the user makes particular sign language movements, the changes in resistance are detected by the sensors, which directly correspond to the bending angles of the fingers.

The sensor outputs are processed by the Arduino Mega 2560. The microcontroller receives the analog signals from the flex sensors and processes them as digital values through its built-in Analog-to-Digital Converter (ADC). The Arduino is initialized with pre-set threshold values corresponding to various finger positions or curvatures. Based on analysis of the sensor values, the Arduino determines the made gesture and assigns it to a particular letter, word, or phrase in a lookup table.

After a gesture has been identified, the translated message is shown on a 16x2 LCD display that is interfaced to the Arduino through the I2C communication interface. The use of the I2C interface reduces the number of wires by utilizing only two data lines, SDA and SCL, to make optimal use of the pins of the Arduino. The system then sends the identified gesture's corresponding text over the SDA line clocked by the SCL line to show on the screen in real-time.

The system runs continuously, scanning data from all five flex sensors and refreshing the output whenever a matching gesture is recognized. To guarantee proper gesture recognition, the sensors are calibrated at the initialization stage to compensate for differences in sensitivity and flexibility. This calibration guarantees that the system accurately recognizes gestures, making it a good and interactive communication device for the hearing and speech impaired.

VI. RESULTS AND DISCUSSIONS

In this work, that was developed to facilitate effective communication, we considered only 6 scenarios. Every scenario has a specific gesture and a specific phrase that denotes a real-life need. The below *Table.1* give an overview about the scenarios included in the 'Handspeak' Library. Here, all the 6 cases gave above 90% success rate.

Test Case	Gesture Detected	Displayed Output	Success Rate
Test 1	No Finger bent	No Gesture	99%
Test 2	Finger 1 bent	Gesture: Food	90%
Test 3	Finger 2 bent	Gesture: Wash Room	95%
Test 4	Finger 3 bent	Gesture: Rest Room	90%
Test 5	Finger 4 bent	Gesture: Class Room	90%
Test 6	Finger 5 bent	Gesture: Friend Ship	95%

Table.1

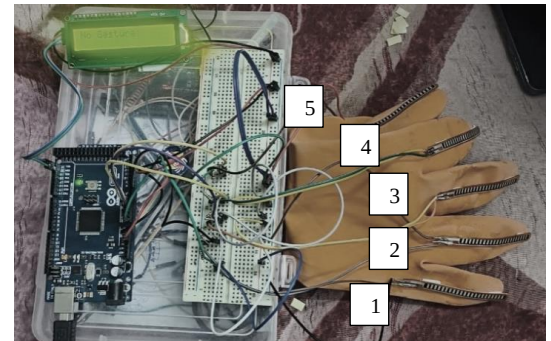


Fig 6: No Flex Sensors Bend

As shown in *Fig. 6*, when the flex sensors are not curved, there is no hand gesture detected by the system. In this case, the screen simply displays "No Gesture" which means that the sensors are in their resting position. This message is used to confirm that the device is powered on and working properly, though no movement has been detected. It's useful because it prevents the system from incorrectly registering a gesture when the hand is just resting, giving a clear indication that no input has been registered. The Results can be further stored in the cloud using various techniques [17].

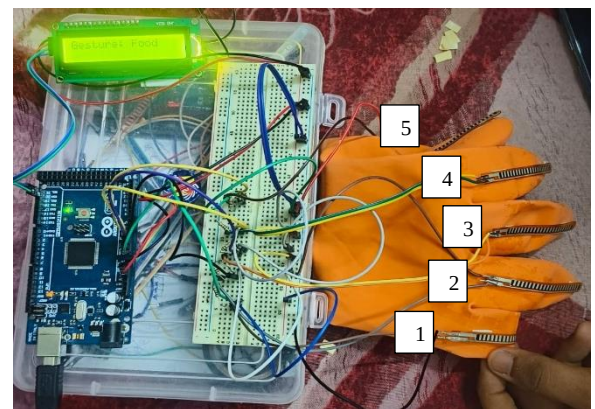


Fig 7: 1st Flex Sensor Bend

For a bend in the first flex sensor, as shown in *Fig. 7*, the system recognizes the specific movement to be a learned gesture. Then, whenever this is the case, the system will

display the screen on the LCD display as "**Gesture: Food.**" This indicates that the system has learned to recognize and detect bending of the first sensor to be the "Food" gesture. The direct and explicit manner in which this data is presented also enables the users to identify that the movement of the hand has been identified and processed by the system correctly.

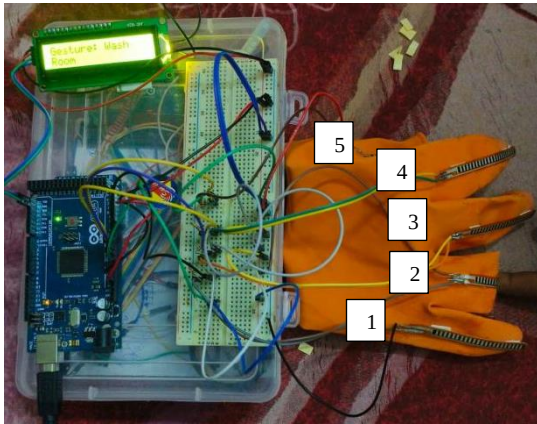


Fig 8: 2nd Flex Sensor Bend

In the above *Fig. 8*, when the second flex sensor is bent, the system recognizes it as another hand gesture. Once more, the LCD display will show "**Gesture: Wash Room,**" which means that the system has correctly recognized the symbol for "Wash Room." This immediate response notifies the user that their movement has been detected and recognized. It facilitates the communication smoother and more naturally and also makes sure that the system is accurately reacting to the sign language gesture of the user.

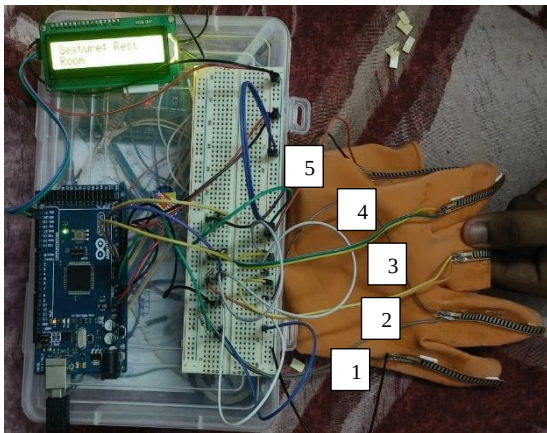


Fig 9: 3rd Flex Sensor Bend

In *Fig. 9*, whenever the third flex sensor is flexed, the system detects this specific hand movement as a new gesture. Here, the LCD screen will show the words "**Gesture: Rest Room,**" which means that the system has properly interpreted the bending of the third sensor as the indication for "Rest Room." This instant visual confirmation on the screen guarantees the user that the gesture has been detected. It serves to make the interaction more intuitive and natural, and it guarantees the system is indeed responding appropriately to the user's sign language gesture and validating that the input has been processed properly.

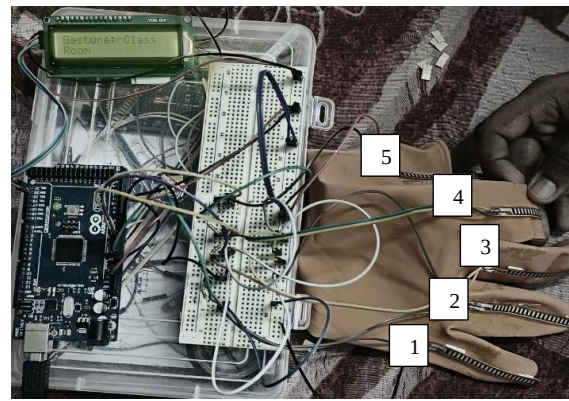


Fig 10: 4th Flex Sensor Bend

As observed in *Fig. 10*, when the fourth flex sensor is flexed, the system recognizes this particular hand movement as another gesture. Here, the LCD screen will show "**Gesture: Class Room,**" which implies that the system has identified correctly the bending of the fourth sensor as the signal for "Class Room." Such a display serves as an instantaneous feedback to the user, and it ensures him that his gesture has been picked up and comprehended. It facilitates making the exchange more natural and fluid, asserting that the system is correctly decoding the user's sign language sign and reacting to it.

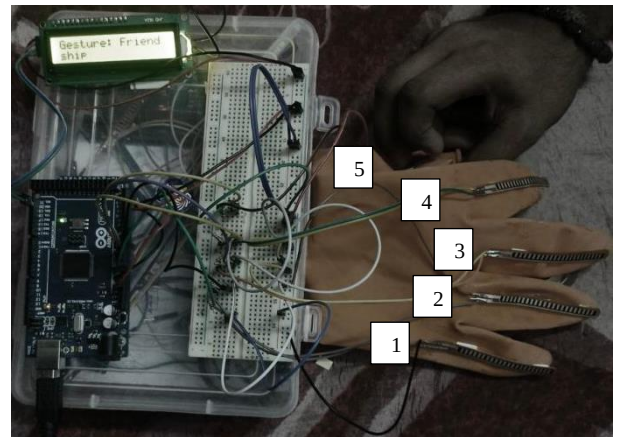


Fig 11: 5th Flex Sensor Bend

As is evident in *Fig. 11*, the system recognizes a specific hand motion when the fifth flex sensor bends. In this case, the LCD display screen will display "**Gesture: Friend Ship,**" specifying that the bending of the fifth sensor has been recognized by the system as the gesture for "Friendship." This immediate display feedback on the screen guarantees to the user that the gesture was recognized and read. It makes the interaction natural and fluid, such that the system is correctly interpreting the user's sign language gesture and responding accordingly, making the communication process more intuitive and efficient.

VII. CONCLUSION

The 'HandSpeak' is an innovative contribution to enhanced communication among differently-abled persons, particularly sign language users. With flex sensors and the technology of Arduino, this device properly captures gestures by the hands and interprets them as readable words or even verbal expressions at the user end, providing an accessible and inexpensive measure. This system can provide ease of everyday communication, empowering differently-abled individuals with hearing or speech loss to interact easily with others.

VIII. FUTURE SCOPE

The 'HandSpeak' has a lot of room to grow. Adding wireless features like Bluetooth, improving gesture recognition with machine learning, and supporting more languages would make it more flexible and user-friendly. Making it wearable and including facial expression recognition could make it feel more natural, while linking it to smart home devices would make everyday tasks easier.

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